Math 10C — Unit 2: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

Part A — Power Vocabulary and Repeated Multiplication

1. State the base and the exponent in each of the following powers.

(a) 7^4 (b) m^{20} (c) 3^n (d) $(-y)^5$ (e) $\left(\frac{2}{9}\right)^7$

2. State the coefficient in each of the following.

(a) $4x^6$ (b) $-8w^3$ (c) b^5 (d) $\frac{t^4}{9}$

3. Write each of the following as a repeated multiplication.

(a) e^4 (b) $5y^3$ (c) $(mn)^2$ (d) $(-6)^3$

(e) p^2q (f) $3\left(\frac{3}{5}\right)^3$ (g) $(5a)^3$ (h) $2cd^2$

(i) $2(cd)^2$ (j) $(2cd)^2$

Part B — Evaluating Powers, Negative Bases and the Zero Exponent

4. Evaluate.

(a) 2^9 (b) -4^2 (c) $(-4)^2$ (d) $(-4)^3$ (e) -4^3 (f) $\left(\frac{5}{8}\right)^3$

5. Evaluate without using a calculator.

(a) -9^2 (b) $(-9)^2$ (c) -9^3 (d) $(-9)^3$

6. Explain why -6^0 and $(-6)^0$ have different values.

7. Evaluate without using a calculator.

(a) 45^0 (b) -1^0 (c) $\left(-\frac{1}{3}\right)^0$ (d) $\frac{1}{3}(9)^0$ (e) $\frac{1}{3}\left(9^2\right)^0$

Part C — The Exponent Laws

8. Write in a simpler form and evaluate.

(a) $5^4 \cdot 5^3$ (b) $\left(6^2\right)^3$ (c) $\frac{9^{18}}{9^{16}}$ (d) $\left(\frac{3}{4}\right)\left(\frac{3}{4}\right)^2$

(e) $\frac{2.5^6}{2.5^4}$ (f) $\left(-2^3\right)^2$ (g) $(-3)^5 \times (-3)^3$ (h) $6^2 \cdot 6^5 \cdot 6^0$

9. Explain, using factors, why $(m^2)(m^4) \neq (m^2)^4$.

10. Use the Product Law to simplify.

(a) $b^5 \times b^3$ (b) $k^7 \cdot k^2$ (c) $(w^4)(w^4)$

11. Use the Quotient Law to simplify.

(a) $\frac{v^9}{v^3}$ (b) $x^8 \div x^5$ (c) $h^9 \div h$

12. Use the Power of a Product Law to simplify.

(a) $(4x)^3$ (b) $\left(\frac{1}{3}c\right)^2$ (c) $(-2mn)^3$

13. Use the Power of a Quotient Law to simplify.

(a) $\left(\frac{m}{n}\right)^3$ (b) $\left(\frac{4}{d}\right)^3$ (c) $\left(\frac{h}{6}\right)^2$

14. Use the Power of a Power Law to simplify.

(a) $(r^3)^4$ (b) $(k^5)^6$ (c) $(t^8)^7$

15. State the value of x in each of the following.

(a) $(a^4)(a^x) = a^{11}$ (b) $b^x \cdot b^5 = b^9$ (c) $c^x \div c^3 = c^{10}$

(d) $\dfrac{d^{12}}{d^x} = d^3$ (e) $(e^x)^4 = e^{20}$ (f) $(f^9)^x = f^9$

16. Use the exponent laws to simplify.

(a) $\dfrac{x^{15}}{x^4}$ (b) $(xy)^9$ (c) $(t^4)^4$ (d) t^4t^4

(e) $y^6 \times y^9$ (f) $\left(\dfrac{a}{b}\right)^{13}$ (g) $\left(\dfrac{d}{2}\right)^5$ (h) $(2st)^7$

17. Find *two* different pairs of whole numbers (m, n) for which $a^m \cdot a^n = (a^m)^n$. Explain, using the exponent laws, how you know your pairs work.

Part D — Multiple Choice and Numerical Response

18. *(Multiple Choice)* Which of the following is equivalent to $(b^4)^3$? A. b^7 B. b^{12} C. b^{64} D. b^{81}

19. *(Numerical Response)* If $(a^n)(a^n)(a^n)(a^n) = a^{24}$, where n is a whole number, then the value of n is _____.

Record your answer left to right. | | | | |

Lesson 2 Assignment *Show your work.*

1. Simplify the following.

(a) $4a^5 \times 6a^3$

(b) $(9b^6)(2b^9)$

(c) $7c^4 \cdot 4c^4$

(d) $(-3x^7)(11x^5)$

(e) $\left(-\frac{1}{3}d^6\right)(-9d^5)$

(f) $0.6e^4 \times 0.5e^2$

2. Simplify.

(a) $15x^7 \div 3x^2$

(b) $(72e^{11}) \div (9e^7)$

(c) $\frac{32d^9}{8d^3}$

(d) $\frac{-54d^{40}}{6d^8}$

(e) $(-24e^{12}) \div (-6e^4)$

(f) $\frac{15f^9}{15f^6}$

3. Write in simplest form.

(a) $(4a^3b^5)(6a^2b^7)$

(b) $x^8y^0x^3y^6$

(c) $\frac{8x^6y^9}{2x^4y^3}$

(d) $\frac{5x^4y^7}{x^3y^2}$

(e) $\frac{4f^{12}g^3}{12f^4g}$

(f) $(7b^4c)(bc^2)(-2b^2c^6)$

4. Simplify.

(a) $\frac{14e^9f^8}{21e^5f^3}$

(b) $(3p^4)(5p^8)(-2p)$

(c) $(-3xy)(x^2y^4)(-5xy)$

(d) $(-9b^7c) \div (3b^3c)$

(e) $(-12t^9y^7) \div (-3t^6y^3)$

(f) $(5x^6z^9) \div (-15xz^7)$

5. Write in simplest form.

(a) $(-a^3b^4)^4$

(b) $(-a^3b^4)^5$

(c) $\left(\frac{c^5}{a^4}\right)^3$

(d) $\frac{c^7 \times c^3}{c^5 \times c^2}$

6. Simplify the following.

(a) $(2ab^3)^5$ (b) $(-3a^4c^3)^4$ (c) $(-3m^4n^5)^4(m^2n^2)$

(d) $(-2x^3y^4)^3(5xy^6)$ (e) $(a^2b^3c^4)(2abc^3)^3$

7. Write each expression in simplest form without brackets.

(a) $\left(\dfrac{3d^4 \times d^5}{6d^2}\right)^3$ (b) $\left(\dfrac{-24a^4b^2 \cdot 3a^3b^5}{12ab^4}\right)^2$

(c) $\left(\dfrac{-4k^4 \cdot k^3}{k}\right)^2 \left(\dfrac{(-k)^6 \cdot k^2}{4k^3}\right)$

8. Write in a simpler form and evaluate.

(a) $\dfrac{5^7 \times 5}{5^3}$ (b) $(-2^5)^2$ (c) $\left(\dfrac{3^8}{3^3}\right)^2$ (d) $\dfrac{(0.6)^9}{(0.6)^5 \times (0.6)^3}$

(e) $-4^5 \times 4^3$ (f) $(-4)^5 \times (-4)^3$ (g) $-9^8 \div (-9)^6$ (h) $\dfrac{-9^8}{-9^6}$

9. Write each expression in simplest form without brackets.

(a) $(-x)^{14} \div (-x)^7$ (b) $(-a)^{11} \div (-a)^6$ (c) $-p^{12} \div (-p)^4$

(d) $c^7 \div (-c)^3$ (e) $-(-t)^5 \div (-t)^4$ (f) $-(-s^6) \div (-(s^5))$

10. *(Multiple Choice)* The simplified form of $\dfrac{1}{49}(3x^2)^2(-7yx^3)$ is

A. $\dfrac{9}{7}x^7y$ B. $-\dfrac{3}{7}x^7y$ C. $-\dfrac{9}{7}x^7y$ D. $-\dfrac{9}{7}x^{12}y$

11. *(Numerical Response)* The expression $(-2m^3n^2)^2(-3m^2n)^3\left(\dfrac{1}{36m^9n^4}\right)$ can be simplified to the form am^bn^c, where a, b, c are integers. The value of $a+b+c$ is _____.

Record your answer left to right. | | | | |

12. *(Multiple Choice)* Which statement is true for all nonzero values of the variables?

A. $(a+b)^2 = a^2+b^2$
B. $2^3 \cdot 3^2 = 6^5$
C. $a^3 + a^2 = a^5$
D. $(2a)^3 \cdot a^2 = 8a^5$

13. *(Extension)* Simplify each expression.

(a) $a^{x+2y} \cdot a^{3x+y}$ (b) $\dfrac{m^{x+11}}{m^5}$ (c) $\dfrac{a^{4m+3}}{a^{m-2}}$ (d) $\dfrac{x^{3y+8} \cdot x^{2y+1}}{x^{y+6}}$

Lesson 3 Assignment *Show your work.*

Complete Questions 1–21, writing every final answer with positive exponents unless the question asks you to evaluate. Questions 18–21 use scientific notation.

1. Write the following with positive exponents.

(a) x^{-4} (b) y^{-6} (c) 5^{-1} (d) $\dfrac{1}{a^{-8}}$ (e) $\dfrac{1}{7^{-2}}$

2. Without using a calculator, show that $4 \div 2^{-3} = 32$.

3. Simplify, express with positive exponents, and evaluate without using a calculator.

(a) $5^2 \times 5^{-5}$ (b) $4^0 \times 4^{-2}$ (c) $\dfrac{1}{6^{-2}}$ (d) $7^{-2} \div 7$ (e) $(4^2)^{-2}$

4. Express with positive exponents.

(a) p^3q^{-4} (b) $c^{-5}x^{-2}$ (c) $25h^{-1}$ (d) $\dfrac{3}{4}b^{-8}$ (e) $(z^{-3})^{-2}$

(f) $\dfrac{s^{-7}}{9}$ (g) $\dfrac{1}{9y^{-5}}$ (h) $\dfrac{5}{y^{-8}}$ (i) $\dfrac{p^3}{q^{-6}}$ (j) $\dfrac{p^{-3}}{q^6}$

5. Evaluate the following without using a calculator.

(a) -4^{-2} (b) $(-4)^{-2}$ (c) $-5^2 \cdot 3^{-2}$ (d) $(-6.2)^0$ (e) $[-(2.7)^0]^{-2}$

6. Use a calculator to find the exact value of the following (see the calculator tip in Section 2).

(a) -3^{-5} (b) $(-6)^{-3}$ (c) $(0.5)^{-3}$ (d) $(-0.02)^{-2}$ (e) $\left(\dfrac{3}{8}\right)^{-3}$

7. State whether the following are true or false.

(a) $8x^{-4} = \dfrac{8}{x^4}$ (b) $3a^{-5} = \dfrac{1}{3a^5}$ (c) $\dfrac{6}{c^{-8}} = 6c^8$ (d) $\dfrac{x^{-3}}{5} = \dfrac{5}{x^3}$

(e) $\dfrac{1}{4y^{-1}} = 4y$ (f) $\dfrac{1}{\frac{1}{3}p} = 3p^{-1}$ (g) $(2x)^4 = \dfrac{1}{(2x)^{-4}}$ (h) $\dfrac{1}{\left(\frac{1}{8}a\right)^{-2}} = 64a^2$

8. Simplify and write the answer with positive exponents.

(a) $x^8 \cdot x^{-3}$ (b) $m^4 \div m^9$ (c) $b^{-2} \cdot b^{-1}$ (d) $-w^0 \div w^3$

9. Simplify and write the answer with positive exponents.

(a) $a^6 \times a^{-9}$ (b) $8x^3 \div 4x^{-2}$ (c) $\dfrac{10y^{-5}}{2y^{-2}}$ (d) $\dfrac{3a^{-4}}{9b^5}$

(e) $-9x^{-3}$ (f) $-(4x)^{-2}$ (g) $(-4x)^{-2}$ (h) $\dfrac{(-4x)^{-2}}{-4x^{-2}}$

10. Simplify each expression, writing the answer with positive exponents.

(a) $a^{-4}a^{-2}$ (b) $(4b^6b^{-10})(-8b^2b^{-9})$ (c) $(-6x^4x^{-6})(x^3x^{-2})$

(d) $(-3a^2)^{-3} \cdot 6a^{15}$ (e) $\dfrac{25a^8b^{-4}}{-5a^8b^5}$ (f) $(-2a^4b^{-2}c^0)^{-2}$

11. Simplify. Write the final answer with positive exponents.

(a) $\dfrac{20a^4b^{-3}}{5a^{-6}b^{-1}} \times \dfrac{-9a^{-2}}{-3b^{-5}}$ (b) $\dfrac{12(p^2q^3r^0)^{-3}}{(6p^{-4}q^2r^2)^{-2}}$

(c) $(-2x^3y^4z^5)^{-2}(-2x^2y^{-6}z^8)^3$ (d) $(6a^3b^2)(-2a^{-3}b)^{-3} \div (-3a^6b^{-5})^{-2}$

12. Evaluate the following without using a calculator.

(a) $\left(\dfrac{3}{4}\right)^{-2}$ (b) $\left(\dfrac{1}{6}\right)^{-2}$ (c) $\left(\dfrac{9}{4}\right)^{-1}$ (d) $\left(\dfrac{5}{3}\right)^{-3}$

13. Simplify. Write the final answers with positive exponents.

(a) $\left(\frac{e}{f}\right)^{-3}$ (b) $\left(\frac{x}{5}\right)^{-3}$ (c) $\left(\frac{p^2}{r^2}\right)^{-3}$

(d) $\left(\frac{a^{-3}}{b^{-4}}\right)^{-2}$ (e) $\left(\frac{-15x^{-2}}{5y^{-6}}\right)^{-1}$ (f) $\left(\frac{15x^3y^{-1}}{-9x^{-3}y^4}\right)^{-2}$

14. Simplify. Write the final answers with positive exponents.

(a) $\left(\frac{-x^5}{y}\right)^{-2} \div \left(\frac{y^4}{x^3}\right)^2$ (b) $25\left(\frac{5w^3x^{-4}z^2}{w^{-2}z}\right)^{-2} \times \frac{20(x^3z^6)^0}{x^{-5}z^7}$

15. *(Identify and correct the error)* Sam simplified $\left(2x^{-3}\right)^{-2}$ as shown.

$$\begin{aligned} \text{Line 1:} \quad \left(2x^{-3}\right)^{-2} &= 2 \cdot x^{(-3)(-2)} \\ \text{Line 2:} \quad &= 2x^6 \end{aligned}$$

(a) Identify the line with the error, and explain the mistake.

(b) Write a correct simplification with positive exponents. Check it with $x = 1$.

16. *(Multiple Choice)* The value of $\frac{1^{-4} + 2^0}{5^{-1}}$ is A. 2 B. 5 C. 10 D. 25

17. *(Multiple Choice)* Which of the following statements are true?

i) $5a^{-4} = \frac{1}{5a^4}$ ii) $12x^5 \div 3x^8 = \frac{1}{4x^3}$ iii) $\frac{1}{3a} = 3a^{-1}$

A. i) only B. ii) only C. iii) only D. none of the statements are true

18. Express each number in scientific notation.
(a) 8 250 000 (b) 0.000041

19. Express 5.62×10^{-8} in standard notation.

20. Evaluate and write each result in scientific notation.
(a) $(2.4 \times 10^{6})(3.5 \times 10^{-2})$ (b) $(4.5 \times 10^{7}) \div (1.5 \times 10^{3})$

21. *(Numerical Response)* The speed of light is 3×10^{8} m/s. A space probe is 2.4×10^{10} m from Earth. If the time, in seconds, for a radio signal to travel this distance is expressed in scientific notation in the form $a \times 10^{n}$, the value of $a + n$ is _____.

Record your answer left to right. | | | | |

Lesson 4 Assignment *Show your work.*

Part A — Evaluating Rational-Exponent Powers

1. Evaluate without the use of a calculator.

(a) $9^{1/2}$ (b) $400^{1/2}$ (c) $216^{1/3}$ (d) $4^{3/2}$

(e) $36^{3/2}$ (f) $81^{3/4}$ (g) $32^{2/5}$ (h) $(0.09)^{0.5}$

2. Determine the exact value without using a calculator.

(a) $16^{-1/2}$ (b) $9^{-5/2}$ (c) $8^{-2/3}$ (d) $100^{-3/2}$

(e) $32^{-3/5}$ (f) $27^{-2/3}$ (g) $\left(13^2 - 12^2\right)^{-1/2}$ (h) $(0.25)^{-3/2}$

3. Determine the exact value without using a calculator.

(a) $\left(\frac{1}{49}\right)^{1/2}$ (b) $\left(\frac{1}{9}\right)^{-1/2}$ (c) $\left(\frac{1}{27}\right)^{4/3}$ (d) $\left(\frac{25}{4}\right)^{-3/2}$ (e) $\left(\frac{81}{16}\right)^{-3/4}$

4. Determine the exact value without using a calculator.

(a) $(-64)^{1/3}$ (b) $(-125)^{2/3}$ (c) $-16^{-1/2}$ (d) $-(-243)^{-3/5}$ (e) $(-0.027)^{2/3}$

5. Use a calculator to evaluate the following to the nearest hundredth.

(a) $5^{2/3}$ (b) $9^{3/5}$ (c) $(-6)^{4/3}$ (d) $8^{-1/5}$ (e) $-(-0.5)^{4/5}$

6. Explain your reasoning.

(a) Explain, using the exponent laws, why $16^{1/2} = 4$ and $16^{1/4} = 2$.

(b) Explain why $(-16)^{1/4}$ has no real value but $(-8)^{1/3}$ does.

7. *(Identify and correct the error)* Priya evaluated $8^{-2/3}$ as shown.

$$\begin{aligned} \text{Line 1:} \quad 8^{-2/3} &= -\left(8^{2/3}\right) \\ \text{Line 2:} \quad &= -\left(\sqrt[3]{8}\right)^2 \\ \text{Line 3:} \quad &= -4 \end{aligned}$$

(a) Identify the line with the error, and explain the mistake.

(b) Write a correct solution.

Part B — Converting Between Powers and Radicals

8. Write an equivalent expression using radicals.

(a) $p^{1/6}$ (b) $q^{1/4}$ (c) $r^{1/7}$ (d) $s^{-1/3}$

(e) $t^{-1/9}$ (f) $u^{5/6}$ (g) $v^{7/4}$ (h) $w^{n/k}$

9. Assuming x represents a positive integer, state which *two* of the following expressions have no meaning.

(a) $(-x)^{5/3}$ (b) $(-x)^{5/4}$ (c) $-(-x)^{1/7}$ (d) $-(-x)^{7/6}$

10. Write each power in radical form.

(a) $p^{2/3}$ (b) $q^{5/2}$ (c) $r^{1/6}$ (d) $s^{-3/4}$ (e) $t^{-1/5}$

(f) $3u^{4/5}$ (g) $(3u)^{4/5}$ (h) $-w^{3/2}$ (i) $(-w)^{3/2}$ $(w < 0)$ (j) $4v^{-1/3}$

11. Write each radical as a power in the form z^n, where $n \in Q$.

(a) $\sqrt[6]{z^5}$ (b) $\sqrt[7]{z^3}$ (c) $\sqrt{z^{11}}$ (d) $\dfrac{1}{\sqrt[5]{z}}$ (e) $\dfrac{1}{\sqrt[5]{z^4}}$

12. In each case, write the given number as a power with the given exponent.

(a) 7 as a power with an exponent of $\frac{1}{2}$ (b) 5 as a power with an exponent of $\frac{1}{4}$ (c) -4 as a power with an exponent of $\frac{1}{3}$

(d) $\frac{1}{9}$ as a power with an exponent of $-\frac{1}{2}$ (e) 8 as a power with an exponent of $-\frac{1}{3}$ (f) 1 000 as a power with an exponent of $\frac{3}{4}$

Part C — Solving Problems with Rational Exponents

13. A cube has a volume of 512 cm^3.
(a) Write a power that represents the edge length of the cube.
(b) Write a power that represents the surface area of the cube.
(c) Calculate the exact edge length and surface area of the cube.

14. A cube has a volume of W cm^3.
(a) Write a power *and* a radical that represent the edge length of the cube.
(b) Write a power *and* a radical that represent the area of one face of the cube.

15. The volume of a sphere is $V = \frac{4}{3}\pi r^3$, so its radius is $r = \left(\frac{3V}{4\pi}\right)^{1/3}$.
(a) Write the radius formula in radical form.
(b) A ball bearing has a volume of 36π mm^3. Find its exact radius.
(c) A weather balloon holds 4 000 m^3 of gas. Find its radius to the nearest tenth of a metre.

16. A bacteria culture starts with 500 cells and doubles every 3 hours, so the number of cells after t hours is $P = 500(2)^{t/3}$.
(a) Write the number of cells after 2 hours using a rational exponent, and then in radical form.
(b) Evaluate your answer to (a) to the nearest whole number.
(c) How many cells are there after 4.5 hours, to the nearest whole number?

17. A square garden has an area of A m^2.
(a) Write the side length as a power and as a radical.
(b) The area is 50 m^2. How much fencing goes all the way around? Answer to the nearest tenth of a metre.
(c) The area is made 4 times as large. Use the exponent laws to show what happens to the side length.

Part D — Multiple Choice and Numerical Response

18. *(Multiple Choice)*
(a) $\left(\frac{16}{81}\right)^{-0.5}$ is equal to
A. $\frac{81}{256}$ B. $\frac{4}{9}$ C. $\frac{9}{4}$ D. $-\frac{4}{9}$

(b) $\left(-\frac{1}{27}\right)^{-4/3}$ is equal to
A. 81 B. -81 C. $\frac{1}{81}$ D. has no meaning

19. *(Numerical Response)* Evaluate the following and arrange the answers from greatest to least.

Calculation 1: $-\left(64^{-2/3}\right)$ Calculation 2: $\left(\frac{1}{64}\right)^{1/3}$

Calculation 3: $(-64)^{2/3}$ Calculation 4: $\left(-\frac{1}{64}\right)^{-1/3}$

Place the calculation number with the greatest answer in the first box, and so on down to the smallest answer in the last box.

Record your answer left to right. | | | | |

Lesson 5 Assignment *Show your work.*

Part A — Simplifying and Converting

1. Simplify each expression. Write your answer as a power with a positive exponent, and then in radical form.

(a) $x^{9/4} \times x$ (b) $y^{7/6} \div y^{5/6}$ (c) $\left(a^{3/7}\right)^{7/2}$ (d) $\left(m^4 n\right)^{2/3}$

(e) $x^{2/3} \times x^{-1}$ (f) $y^{3/8} \div y^{7/8}$ (g) $\left(\dfrac{p}{q^3}\right)^{1/2}$ (h) $\left(\dfrac{p^2}{q}\right)^{-5/2}$

2. Write each expression as a single power, then evaluate.

(a) $\sqrt{\sqrt[3]{4096}}$ (b) $\dfrac{1}{\sqrt[4]{81}}$ (c) $\sqrt{\sqrt{6561}}$

3. Write each expression in the form ax^n, where $a \in I$ and $n \in Q$.

(a) $\sqrt[3]{27x^{10}}$ (b) $\sqrt[5]{243x^{20}}$ (c) $\sqrt{144x}$

(d) $\left(\sqrt[4]{x^9}\right)\left(\sqrt[4]{x^3}\right)$ (e) $\sqrt[3]{-125x}$ (f) $4\sqrt[3]{x} \times 2\sqrt[4]{x}$

4. Write an equivalent expression using positive rational exponents.

(a) $\sqrt{\sqrt{z^9}}$ (b) $\sqrt[3]{\sqrt{a^{10}}}$ (c) $\sqrt[3]{\sqrt{15\,625y^{18}}}$ (d) $\sqrt[4]{\sqrt[3]{z^5}}$

(e) $\left(\sqrt[5]{3w-2}\right)^{-2}$ (f) $\left(\sqrt[5]{z^5 w^{10}}\right)^{4/3}$ (g) $-\sqrt[5]{z^3}$ (h) $\sqrt[5]{(-z)^3}$

Part B — Multi-Law Chains and Error Correction

5. Simplify. Write each answer with positive exponents.

(a) $\left(16x^{-4}y^{6}\right)^{-3/4}$

(b) $\left(27a^{1/2}b^{-3}\right)^{2/3} \div \left(9a^{-1}\right)^{1/2}$

(c) $4x^{1/3} \cdot (8x)^{-2/3}$

(d) $\left(25m^{-4}n^{2/3}\right)^{-1/2} \cdot 5m^{-1}$

6. *(Identify and correct the error)* Malik simplified $\left(x^{1/2}\right)^{4} \cdot x^{1/2}$ as shown.

$$\begin{aligned} \text{Line 1:} \quad & \left(x^{1/2}\right)^{4} \cdot x^{1/2} = x^{\frac{1}{2}+4} \cdot x^{1/2} \\ \text{Line 2:} \quad & = x^{9/2} \cdot x^{1/2} \\ \text{Line 3:} \quad & = x^{5} \end{aligned}$$

(a) Identify the line with the error, and explain the mistake.

(b) Write a correct simplification.

Part C — Matching, Multiple Choice and Numerical Response

7. *(Matching)* Match each item in Column 1 with the equivalent item in Column 2. Each item in Column 2 may be used once, more than once, or not at all.

Column 1

(a) $\left(\frac{p}{q}\right)^{5/2}$ (b) $\left(\frac{p}{q}\right)^{2/5}$ (c) $\left(\frac{q}{p}\right)^{-5/2}$

(d) $\left(\frac{p}{q}\right)^{-2/5}$ (e) $\left(\frac{q}{p}\right)^{2/5}$ (f) $\left(\frac{p}{q}\right)^{-5/2}$

Column 2

A. $\sqrt[5]{\frac{q^2}{p^2}}$ B. $\sqrt{\frac{p^2}{q^2}}$ C. $-\sqrt[5]{\frac{p^2}{q^2}}$

D. $\sqrt{\frac{p^5}{q^5}}$ E. $\sqrt{\frac{q^5}{p^5}}$ F. $-\sqrt{\frac{q^5}{p^5}}$

(a) ____ (b) ____ (c) ____ (d) ____ (e) ____ (f) ____

8. *(Multiple Choice)* Which of the following is equivalent to $(-x^5)^{-4/5}$?

A. x^4 B. $-x^4$ C. $\frac{1}{x^4}$ D. $-\frac{1}{x^4}$

9. *(Multiple Choice)* Which expression is *not* equivalent to the others?

A. $b^{-3/5}$ B. $\left(\frac{1}{b^5}\right)^{3}$ C. $\left(\sqrt[5]{b}\right)^{-3}$ D. $\frac{1}{\sqrt[5]{b^3}}$

10. *(Multiple Choice)* For all positive integers c and d, which of the following is *not* equivalent to $c^2\sqrt[3]{d}$?

A. $c^2d^{1/3}$

B. $\left(c^6d\right)^{1/3}$

C. $\sqrt[3]{c^6d}$

D. All of the expressions are equivalent to $c^2\sqrt[3]{d}$.

11. *(Numerical Response)* Find the value, to the nearest whole number, of $\left(x^{2/3}-y^{1/2}+z^{1/4}\right)^2$ when $x=64$, $y=50$, and $z=81$.

Record your answer left to right. | | | | |

Lesson 6 Assignment *Show your work.*

Part A — Vocabulary and Repeated Multiplication

1. *(Multiple Choice)* Which statement is true? (Assume $c \neq 0$.)

A. In $(-5)^3$, the base is 5.
B. The coefficient of $-\dfrac{3x^4}{7}$ is -3.
C. $-c^0 = -1$
D. $(-c)^0 = -1$

2. *(Multiple Choice)* Consider the following two statements about repeated multiplication.

Statement 1: $4mn^5 = 4 \times m \times n \times n \times n \times n \times n$ Statement 2: $7(pq)^2 = 7 \times 7 \times p \times p \times q \times q$

Which one of the following is true?
A. Statement 1 is correct and Statement 2 is incorrect. B. Statement 2 is correct and Statement 1 is incorrect.
C. Both statements are correct. D. Both statements are incorrect.

3. *(Multiple Choice)* Which of the following can be simplified to c^{10}?

I. $c^6 \times c^4$ II. $c^{12} + c^2$ III. $c^{16} \div c^6$ IV. $c^5 + c^5$

A. I and III only B. II and IV only C. I, II, III, and IV D. none of I, II, III, and IV

Part B — Product and Quotient Laws

4. *(Multiple Choice)* $7b^3 \times 4b^6$ can be simplified to A. $11b^{18}$ B. $28b^9$ C. $28b^{18}$ D. $11b^9$

5. *(Multiple Choice)* $\dfrac{12x^{18}}{4x^5}$ can be simplified to A. $3x^{13}$ B. $3x^{23}$ C. $8x^{13}$ D. $8x^{23}$

6. *Use the following information to answer this question.*

$(-3p^2q)(-2pq^4)(-5q^3)^2$ can be written in the form ap^xq^y, where $a, x,$ and y are integers.

(a) *(Multiple Choice)* The value of a is
A. 150 B. -150 C. 30 D. -30

(b) *(Numerical Response)* The value of $x + y$ is _____.

Record your answer left to right. | | | | |

Part C — Negative Exponents

7. *(Multiple Choice)* $\dfrac{1}{k^{-3}}$ is equivalent to A. k^3 B. $\dfrac{1}{k^3}$ C. $-3k$ D. $-k^3$

8. *(Multiple Choice)* $\dfrac{10x^2}{5x^{-6}}$ can be simplified to A. $2x^{-4}$ B. $2x^8$ C. $5x^8$ D. $5x^{-4}$

9. *(Multiple Choice)* $6y^{-3}$ is equivalent to A. $\dfrac{1}{6y^3}$ B. $-18y$ C. $\dfrac{1}{36y^3}$ D. $\dfrac{6}{y^3}$

Part D — Rational Exponents

10. *(Multiple Choice)* Expressed in radical form, $x^{2/9}$ is equivalent to
A. $\sqrt[2]{x^9}$ B. $9\sqrt{x^2}$ C. $\sqrt[9]{x^2}$ D. $\dfrac{1}{9}\sqrt{x^2}$

11. *(Numerical Response)* If $p = 16$ and $q = -8$, the value of $p^{3/2} - q^{-2/3}$ to the nearest tenth is _____.
Record your answer left to right. ☐☐☐☐

12. *(Numerical Response)* $\left(\sqrt[3]{a^2}\right)\left(\sqrt{a^5}\right)$ can be written in the form $a^{p/6}$. The value of p is _____.
Record your answer left to right. ☐☐☐☐

13. *(Multiple Choice)* $\left(27x^6\right)^{2/3}$ simplifies to
A. $9x^4$ B. $18x^4$ C. $9x^9$ D. $27x^4$

14. *(Multiple Choice)* $x^{1/2} \cdot x^{1/3} \div x^{-1/6}$ simplifies to
A. $x^{2/3}$ B. x C. $x^{1/6}$ D. $x^{5/6}$

15. *(Numerical Response)* $\left(4a^{-2/3}\right)^{3/2} \cdot 2a^2$ can be written in the form ka^n. The value of $k + n$ is _____.
Record your answer left to right. ☐☐☐☐

Part E — Written Response

16. **Written Response — 5 marks** Jordan simplified $\left(27a^3b^{-6}\right)^{-2/3} \cdot \left(a^{1/2}\right)^6$ as shown.

$$\begin{aligned}
\text{Step 1:} \quad \left(27a^3b^{-6}\right)^{-2/3} \cdot \left(a^{1/2}\right)^6 &= 27^{-2/3}a^{-2}b^4 \cdot a^3 \\
\text{Step 2:} \quad &= -18a^{-2}b^4 \cdot a^3 \\
\text{Step 3:} \quad &= -18a^{-6}b^4 \\
\text{Step 4:} \quad &= -\frac{18b^4}{a^6}
\end{aligned}$$

(a) Step 1 is correct. Name the exponent law(s) Jordan used in Step 1. *(1 mark)*

(b) Identify the error in Step 2 and the error in Step 3, and explain each mistake. *(2 marks)*

(c) Write a correct simplification with positive exponents. *(1 mark)*

(d) Verify your answer by substituting $a = 1$ and $b = 1$ into the original expression and into your answer. *(1 mark)*

Part F — Scientific Notation

17. *(Multiple Choice)* If $(2.4 \times 10^{-2}) \times (5.5 \times 10^5) = 1.32 \times 10^n$, then the value of n is

A. 3 B. 4 C. 5 D. 6

18. *(Multiple Choice)* A sheet of gold leaf is 9.2×10^{-6} m thick. Approximately how many sheets are required to make a stack 0.46 m high?

A. 5 000 B. 50 000 C. 500 000 D. 5 000 000

19. *Use the following information to answer this question.*

The average height of an adult giraffe is 5.5 m. Human hair grows about 1.20×10^{-2} m per month. The current population of India is approximately 1.40×10^{9}. The circumference of the Moon is about 1.09×10^{7} m.	The mass of Jupiter is about 1.90×10^{27} kg. The mass of the Earth is about 5.98×10^{24} kg. The mass of a proton is approximately 1.67×10^{-27} kg.

(a) Determine, to the nearest million, the number of giraffes that would need to lie head-to-hoof (each one 5.5 m long) to encircle the Moon.

(b) Determine an estimate for the total length of hair growth in one month, summed across the entire population of India. Answer in scientific notation to the nearest hundredth.

(c) Approximately how many protons would have the same combined mass as the planet Jupiter? Answer in scientific notation to the nearest hundredth.

(d) How many times heavier is Jupiter than the Earth? Answer in standard decimal form to the nearest whole number.

Answer Key — Lesson Assignments

Lesson 1 Assignment

1. (a) base 7, exp. 4 (b) base m, exp. 20 (c) base 3, exp. n (d) base $-y$, exp. 5 (e) base $\frac{2}{9}$, exp. 7.
2. (a) 4 (b) -8 (c) 1 (d) $\frac{1}{9}$.
3. (a) $e \times e \times e \times e$ (b) $5 \times y \times y \times y$ (c) $m \times n \times m \times n$ (d) $(-6) \times (-6) \times (-6)$ (e) $p \times p \times q$ (f) $3 \times \frac{3}{5} \times \frac{3}{5} \times \frac{3}{5}$ (g) $5 \times a \times 5 \times a \times 5 \times a$ (h) $2 \times c \times d \times d$ (i) $2 \times c \times d \times c \times d$ (j) $2 \times c \times d \times 2 \times c \times d$.
4. (a) 512 (b) -16 (c) 16 (d) -64 (e) -64 (f) $\frac{125}{512}$.
5. (a) -81 (b) 81 (c) -729 (d) -729.
6. $-6^0 = -1$ (the exponent applies only to 6); $(-6)^0 = 1$ (the exponent applies to the base -6).
7. (a) 1 (b) -1 (c) 1 (d) $\frac{1}{3}$ (e) $\frac{1}{3}$.
8. (a) $5^7 = 78\,125$ (b) $6^6 = 46\,656$ (c) $9^2 = 81$ (d) $\left(\frac{3}{4}\right)^3 = \frac{27}{64}$ (e) $2.5^2 = 6.25$ (f) $(-8)^2 = 64$ (g) $(-3)^8 = 6\,561$ (h) $6^7 = 279\,936$.
9. $(m^2)(m^4) = m^6$ has six factors of m, while $(m^2)^4 = m^8$ has eight, so the two are not equal in general.
10. (a) b^8 (b) k^9 (c) w^8.
11. (a) v^6 (b) x^3 (c) h^8.
12. (a) $64x^3$ (b) $\frac{1}{9}c^2$ (c) $-8m^3n^3$.
13. (a) $\frac{m^3}{n^3}$ (b) $\frac{64}{d^3}$ (c) $\frac{h^2}{36}$.
14. (a) r^{12} (b) k^{30} (c) t^{56}.
15. (a) $x = 7$ (b) $x = 4$ (c) $x = 13$ (d) $x = 9$ (e) $x = 5$ (f) $x = 1$.
16. (a) x^{11} (b) x^9y^9 (c) t^{16} (d) t^8 (e) y^{15} (f) $\frac{a^{13}}{b^{13}}$ (g) $\frac{d^5}{32}$ (h) $128s^7t^7$.
17. $(m, n) = (2, 2)$ and $(0, 0)$: both make $m + n = mn$, so $a^{m+n} = a^{mn}$.
18. B.
19. 6.

Lesson 2 Assignment

1. (a) $24a^8$ (b) $18b^{15}$ (c) $28c^8$ (d) $-33x^{12}$ (e) $3d^{11}$ (f) $0.3e^6$.
2. (a) $5x^5$ (b) $8e^4$ (c) $4d^6$ (d) $-9d^{32}$ (e) $4e^8$ (f) f^3.
3. (a) $24a^5b^{12}$ (b) $x^{11}y^6$ (c) $4x^2y^6$ (d) $5xy^5$ (e) $\frac{1}{3}f^8g^2$ (f) $-14b^7c^9$.
4. (a) $\frac{2}{3}e^4f^5$ (b) $-30p^{13}$ (c) $15x^4y^6$ (d) $-3b^4$ (e) $4t^3y^4$ (f) $-\frac{1}{3}x^5z^2$.
5. (a) $a^{12}b^{16}$ (b) $-a^{15}b^{20}$ (c) $\dfrac{c^{15}}{a^{12}}$ (d) c^3.
6. (a) $32a^5b^{15}$ (b) $81a^{16}c^{12}$ (c) $81m^{18}n^{22}$ (d) $-40x^{10}y^{18}$ (e) $8a^5b^6c^{13}$.
7. (a) $\frac{1}{8}d^{21}$ (b) $36a^{12}b^6$ (c) $4k^{17}$.
8. (a) 3 125 (b) 1 024 (c) 59 049 (d) 0.6 (e) $-65\,536$ (f) 65 536 (g) -81 (h) 81.
9. (a) $-x^7$ (b) $-a^5$ (c) $-p^8$ (d) $-c^4$ (e) t (f) $-s$.
10. C.
11. 3.
12. D.
13. (a) a^{4x+3y} (b) m^{x+6} (c) a^{3m+5} (d) x^{4y+3}.

Lesson 3 Assignment

1. (a) $\frac{1}{x^4}$ (b) $\frac{1}{y^6}$ (c) $\frac{1}{5}$ (d) a^8 (e) 7^2.
2. $4 \div 2^{-3} = 4 \times 2^3 = 4 \times 8 = 32$.
3. (a) $\frac{1}{125}$ (b) $\frac{1}{16}$ (c) 36 (d) $\frac{1}{343}$ (e) $\frac{1}{256}$.
4. (a) $\frac{p^3}{q^4}$ (b) $\frac{1}{c^5x^2}$ (c) $\frac{25}{h}$ (d) $\frac{3}{4b^8}$ (e) z^6 (f) $\frac{1}{9s^7}$ (g) $\frac{y^5}{9}$ (h) $5y^8$ (i) p^3q^6 (j) $\frac{1}{p^3q^6}$.
5. (a) $-\frac{1}{16}$ (b) $\frac{1}{16}$ (c) $-\frac{25}{9}$ (d) 1 (e) 1.
6. (a) $-\frac{1}{243}$ (b) $-\frac{1}{216}$ (c) 8 (d) 2500 (e) $\frac{512}{27}$.
7. (a) T (b) F (c) T (d) F (e) F (f) T (g) T (h) F.
8. (a) x^5 (b) $\frac{1}{m^5}$ (c) $\frac{1}{b^3}$ (d) $-\frac{1}{w^3}$.
9. (a) $\frac{1}{a^3}$ (b) $2x^5$ (c) $\frac{5}{y^3}$ (d) $\frac{1}{3a^4b^5}$ (e) $-\frac{9}{x^3}$ (f) $-\frac{1}{16x^2}$ (g) $\frac{1}{16x^2}$ (h) $-\frac{1}{64}$.
10. (a) $\frac{1}{a^6}$ (b) $-\frac{32}{b^{11}}$ (c) $-\frac{6}{x}$ (d) $-\frac{2a^9}{9}$ (e) $-\frac{5}{b^9}$ (f) $\frac{b^4}{4a^8}$.
11. (a) $12a^8b^3$ (b) $\frac{432r^4}{p^{14}q^5}$ (c) $-\frac{2z^{14}}{y^{26}}$ (d) $-\frac{27a^{24}}{4b^{11}}$.
12. (a) $\frac{16}{9}$ (b) 36 (c) $\frac{4}{9}$ (d) $\frac{27}{125}$.
13. (a) $\frac{f^3}{e^3}$ (b) $\frac{125}{x^3}$ (c) $\frac{r^6}{p^6}$ (d) $\frac{a^6}{b^8}$ (e) $-\frac{x^2}{3y^6}$ (f) $\frac{9y^{10}}{25x^{12}}$.
14. (a) $\frac{1}{x^4y^6}$ (b) $\frac{20x^{13}}{w^{10}z^9}$.
15. (a) Line 1: the coefficient 2 was not raised to the power -2 (b) $\frac{x^6}{4}$.
16. C.
17. D.
18. (a) 8.25×10^6 (b) 4.1×10^{-5}.
19. 0.0000000562.
20. (a) 8.4×10^4 (b) 3×10^4.
21. 9.

Lesson 4 Assignment

1. (a) 3 (b) 20 (c) 6 (d) 8 (e) 216 (f) 27 (g) 4 (h) 0.3.
2. (a) $\frac{1}{4}$ (b) $\frac{1}{243}$ (c) $\frac{1}{4}$ (d) $\frac{1}{1\,000}$ (e) $\frac{1}{8}$ (f) $\frac{1}{9}$ (g) $\frac{1}{5}$ (h) 8.
3. (a) $\frac{1}{7}$ (b) 3 (c) $\frac{1}{81}$ (d) $\frac{8}{125}$ (e) $\frac{8}{27}$.
4. (a) -4 (b) 25 (c) $-\frac{1}{4}$ (d) $\frac{1}{27}$ (e) 0.09.
5. (a) 2.92 (b) 3.74 (c) 10.90 (d) 0.66 (e) -0.57.
6. (a) $16^{1/2} \cdot 16^{1/2} = 16^1$, so $16^{1/2} = 4$; $16^{1/4} = (16^{1/2})^{1/2} = 4^{1/2} = 2$ (b) no real number has a 4th power of -16 (even powers are ≥ 0); $(-2)^3 = -8$, so

$(-8)^{1/3} = -2$.

7. (a) Line 1: a negative exponent means reciprocal, not negative (b) $\frac{1}{4}$.

8. (a) $\sqrt[6]{p}$ (b) $\sqrt[4]{q}$ (c) $\sqrt[7]{r}$ (d) $\frac{1}{\sqrt[3]{s}}$ (e) $\frac{1}{\sqrt[9]{t}}$ (f) $\left(\sqrt[6]{u}\right)^5$ (g) $\left(\sqrt[4]{v}\right)^7$ (h) $\left(\sqrt[k]{w}\right)^n$.

9. (b) and (d) have no meaning.

10. (a) $\sqrt[3]{p^2}$ (b) $\sqrt{q^5}$ (c) $\sqrt[6]{r}$ (d) $\frac{1}{\sqrt[4]{s^3}}$ (e) $\frac{1}{\sqrt[5]{t}}$ (f) $3\sqrt[5]{u^4}$ (g) $\sqrt[5]{81u^4}$ (h) $-\sqrt{w^3}$ (i) $\sqrt{(-w)^3}$ (j) $\frac{4}{\sqrt[3]{v}}$.

11. (a) $z^{5/6}$ (b) $z^{3/7}$ (c) $z^{11/2}$ (d) $z^{-1/5}$ (e) $z^{-4/5}$.

12. (a) $49^{1/2}$ (b) $625^{1/4}$ (c) $(-64)^{1/3}$ (d) $81^{-1/2}$ (e) $\left(\frac{1}{512}\right)^{-1/3}$ (f) $10\,000^{3/4}$.

13. (a) $512^{1/3}$ (b) $6\left(512^{1/3}\right)^2$ (c) edge $= 8$ cm, surface area $= 384$ cm^2.

14. (a) $W^{1/3} = \sqrt[3]{W}$ (b) $W^{2/3} = \left(\sqrt[3]{W}\right)^2$.

15. (a) $r = \sqrt[3]{\frac{3V}{4\pi}}$ (b) 3 mm (c) 9.8 m.

16. (a) $500(2)^{2/3} = 500\sqrt[3]{4}$ (b) 794 cells (c) 1 414 cells.

17. (a) $A^{1/2} = \sqrt{A}$ (b) 28.3 m (c) $(4A)^{1/2} = 2A^{1/2}$: the side doubles.

18. (a) C (b) A.

19. 3214.

Lesson 5 Assignment

1. (a) $x^{13/4} = \sqrt[4]{x^{13}}$ (b) $y^{1/3} = \sqrt[3]{y}$ (c) $a^{3/2} = \sqrt{a^3}$ (d) $m^{8/3}n^{2/3} = \sqrt[3]{m^8n^2}$ (e) $x^{-1/3} = \frac{1}{\sqrt[3]{x}}$ (f) $y^{-1/2} = \frac{1}{\sqrt{y}}$ (g) $\frac{p^{1/2}}{q^{3/2}} = \frac{\sqrt{p}}{\sqrt{q^3}}$ (h) $\frac{q^{5/2}}{p^5} = \frac{\sqrt{q^5}}{p^5}$.

2. (a) $4\,096^{1/6} = 4$ (b) $81^{-1/4} = \frac{1}{3}$ (c) $6\,561^{1/4} = 9$.

3. (a) $3x^{10/3}$ (b) $3x^4$ (c) $12x^{1/2}$ (d) x^3 (e) $-5x^{1/3}$ (f) $8x^{7/12}$.

4. (a) $z^{9/4}$ (b) $a^{5/3}$ (c) $5y^3$ (d) $z^{5/12}$ (e) $\frac{1}{(3w-2)^{2/5}}$ (f) $z^{4/3}w^{8/3}$ (g) $-z^{3/5}$ (h) $(-z)^{3/5}$.

5. (a) $\frac{x^3}{8y^{9/2}}$ (b) $\frac{3a^{5/6}}{b^2}$ (c) $\frac{1}{x^{1/3}}$ (d) $\frac{m}{n^{1/3}}$.

6. (a) Line 1: a power of a power multiplies the exponents (b) $x^{5/2}$.

7. (a) D (b) B (c) D (d) A (e) A (f) E.

8. C.

9. B.

10. D.

11. 142.

Lesson 6 Assignment

1. C.

2. A.

3. A.

4. B.

5. A.

6. (a) A (b) 14.

7. A.

8. B.

9. D.

10. C.

11. 63.8.

12. 19.

13. A.

14. B.

15. 17.

16. (a) Power of a Product and Power of a Power Laws (b) Step 2: $27^{-2/3} = \frac{1}{9}$, not -18; Step 3: $a^{-2} \cdot a^3 = a^1$ (add exponents, don't multiply) (c) $\frac{ab^4}{9}$ (d) both equal $\frac{1}{9}$ at $a = b = 1$.

17. B.

18. B.

19. (a) 2 000 000 giraffes (b) 1.68×10^7 m (c) 1.14×10^{54} protons (d) 318 times.